OOP with Intelligent Systems

DS205.3 – Data Science in Python

Mr. Anton Jayakody

MSc (CS)[R], BSc (CS)

Recommended Reading

- Lawson, B. (2024) Building AI Applications with Python: LLMs, RAG, and Agents. Sebastopol: O'Reilly Media.
- McKinney, W. (2022) Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter. 3rd edn. Sebastopol: O'Reilly Media.

Overview

- AI applications are no longer just static scripts. They are Agents that need memory and tools.
- We model agents using Classes because agents have state (memory) and capabilities (tools).

Python OOP syntax

Concept	Python Implementation
Class/Object	class Name:
State	self.attribute
Behavior	def method(self):
Encapsulation	self._hidden
Inheritance	class Child(Parent):
Composition	self.tool = Tool()

Exercise

- Use the Code in https://github.com/tobijay666/DS205.3Practicals, to continue the activity.
- Create a main function to execute the CalculatorTool().
- Create a TranslationTool class that also implements an execute method (returning "Translated: [task]").
- Modify the Agent.run() method so that it prints the tool's name before executing the task.

Tools in Agents

- An Agent needs to be portable. We inject tools into the Agent at startup.
- If you have 100 different tools, you still only have one Agent class. You simply swap the object passed into __init__.

- Example:

```
class EmailTool:
    def execute(self, message: str) → str:
        …
        return f"Email sent: {message}"
```